# BridgeFinder
# Finding Research Collaborators Across Communities at UIUC

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## Abstract

Students who want research experience often know an instructor but few researchers outside that instructor's circle. BridgeFinder helps them find related researchers and see who might connect them. We have built a three-screen prototype, analyzed UIUC publications from 2015 to 2024, and gathered feedback from four students. Our historical test finds that BridgeFinder predicts more later collaborations than topic-only ranking but fewer than triadic closure. We therefore plan to test whether students find its suggestions relevant, unfamiliar, and useful for contacting someone.

## 1 Objective

A student who enjoyed a computing class might want to apply those skills in biology or agriculture. Searching for papers can reveal relevant names, but it does not tell the student whom they could approach or how to start a conversation.

BridgeFinder helps UIUC undergraduate and early-stage graduate students explore research beyond their existing contacts. A student chooses a familiar researcher, sees suggestions outside that person's collaboration group, and opens a card with topics, papers, and a possible introduction route. The starting researcher's work stands in for the student's interests, so the student needs no publications of their own.

The challenge is to suggest someone relevant who also extends the student's usual circle. The prototype supports this sequence. We still need to test whether the suggestions help students choose someone to contact.

## 2 Related Work

**Researcher search.** Illinois Experts [12] collects institutional profiles and publications, helping students check expertise. A profile helps assess a known researcher, while our task asks whom else to approach from a familiar person's network. Google Scholar [9] supports following topics through papers and author profiles. Those records still leave the student to work out a possible route of contact. BridgeFinder combines the research match with that route.

**Literature maps.** Connected Papers [10] gives a visual overview of related papers, useful when entering an unfamiliar area. Its map describes relationships between papers. ResearchRabbit [11] supports continued exploration from paper collections, but our contact task requires a route between people. We adapt visual exploration to researchers and possible introductions.

**Link prediction.** Liben-Nowell and Kleinberg [5] compare ways to predict future links from network structure. Shared-contact counts provide a simple, explainable baseline, but favor people already nearby. We compare with this triadic-closure baseline to test whether our suggestions beyond shared contacts offer useful alternatives.

**Cross-domain recommendation.** Tang et al. [8] combine research interests and relationships to recommend collaborators across domains. This addresses cross-domain relevance, but ranking alone does not show whether a student understands or can use a suggested introduction. We therefore evaluate perceived usefulness as well as historical prediction.

**Network theory.** Granovetter [4] explains how weak ties can carry information beyond close circles. Burt [2] explains the advantage of connecting separate groups. These ideas motivate looking beyond familiar collaborators. Neither establishes that an observed co-authorship route will lead to a useful introduction for a student.

## 3 Course Concepts and Network Analysis

### 3.1 Graph and communities

Brokerage involves connecting people across groups that have few ties to one another. We use this course concept to suggest researchers outside a familiar person's collaboration group. We use OpenAlex [7] publications from 2015–2024. Nodes are authors affiliated with UIUC on a collected paper. An edge joins two authors who wrote together, weighted by their number of joint papers. We retain papers with at least two UIUC authors and omit external collaborators.

Our dataset contains 49,651 papers that meet this rule, retained from an exhausted harvest of 102,947 OpenAlex works matching our query. OpenAlex is mutable and does not index every real-world paper. Table 1 describes the 2015–2019 graph used to make recommendations.

**Table 1.** Co-authorship graphs from 2015–2019 papers.

| Measure | Dataset |
|---|---|
| Authors | 19,389 |
| Edges | 78,281 |
| Communities | 842 |
| Modularity | 0.884 |

We use Louvain [1] to find communities with many collaborations inside each group. It seeks high *modularity*, meaning more within-group edge weight than expected from node strengths. Louvain avoids the repeated shortest-path calculations used by Girvan–Newman [3], making it a practical choice for this graph.

The high modularity suggests concentrated collaboration groups, making cross-group discovery worth exploring. Disconnected components also raise modularity, so this is not evidence of disciplinary separation. Communities are numbered groups, not verified departments.

### 3.2 Choosing suggestions

Following the proposal, candidates must be in another community and at least three hops from the starting researcher. One hop is one co-authorship edge. This excludes existing collaborators and people who already share a co-author. Candidates must also be reachable so we can show a route. All 500 displayed suggestions across the demo's 50 starting researchers have three-edge routes.

The ranking combines topic overlap with a bonus for introduction paths, as shown below.

$$\text{Score}(u, v) = S(u, v) + 0.5\frac{c_3(u, v)}{1 + c_3(u, v)}. \qquad (1)$$

$S$ is the number of shared topics divided by the number of distinct topics across both researchers. Each researcher has up to ten topics from their 2015–2019 papers. $c_3$ counts three-edge paths through two distinct intermediaries. The bonus grows with the number of routes but stays below 0.5. More distant candidates remain eligible without that bonus. The coefficient and saturation are fixed implementation choices, not learned probabilities.

Without the network, the app would lose its group boundary, distance rule, and introduction routes and become a topic search tool. The resulting suggestions extend beyond shared contacts, although the graph does not tell us whether an introduction would be welcome.

## 4 Design and Implementation

The prototype takes the student through three views.

**1. Researchers.** The student searches 50 starting researchers by name or topic. We chose highly connected authors across large communities so the demo has enough suggestions. The landing page explains the steps and defines a researcher card. As in profile search [12], it provides an entry point without requiring a student profile.

**2. Network.** A map shows the selected researcher, suggestions, and intermediaries. Solid lines show observed co-authorships. Orange dashed lines show suggested connections. This borrows visual exploration from literature maps [10, 11] while focusing on people.

**3. Suggestions.** Selecting a name or dashed line opens a card with an observed shortest route, topics, and example papers. The route appears first in response to student feedback. An optional explanation shows the ranking score. Students can edit, copy, or download an outreach draft. The app does not send email. Figure 1 shows the map and route detail.

The app, historical test, and study cards share one ranking implementation. Recommendations are computed in Python and displayed in D3. The core flow works on desktop and mobile, including keyboard navigation, and has been browser-tested for all 50 starting researchers.

## 5 Users and Feedback

Our target users want research opportunities but have few contacts beyond their classes or department. We gathered feedback from four students as they talked through their use of the prototype. The two graduate students were a Computer Science student and a Master of Computer Science student researching computational biology. One undergraduate studied Electrical and Computer Engineering, and the other studied Food Systems.

The undergraduates match our student entry point, while the graduate students add perspectives from computational research and its applications. We used four sessions to identify problems in the core flow before a broader study [6]. This round does not estimate usability across campus. We made four changes in response to the feedback.

**Getting started.** One student trying to find a relevant professor did not understand that they first needed to select a researcher to enter the graph. We added instructions on the landing page explaining that students should choose a starting researcher, explore a suggestion, and then review its route. The page now explains that the starting person's publications guide the suggestions.

**Understanding the interface.** Students found the interface unintuitive, prompting an overhaul with simpler layouts and clearly named views. They were also unsure what a card meant. The landing page now defines a researcher card as a profile with topics, example papers, and a possible introduction route. The map legend distinguishes observed collaborations from suggestions.

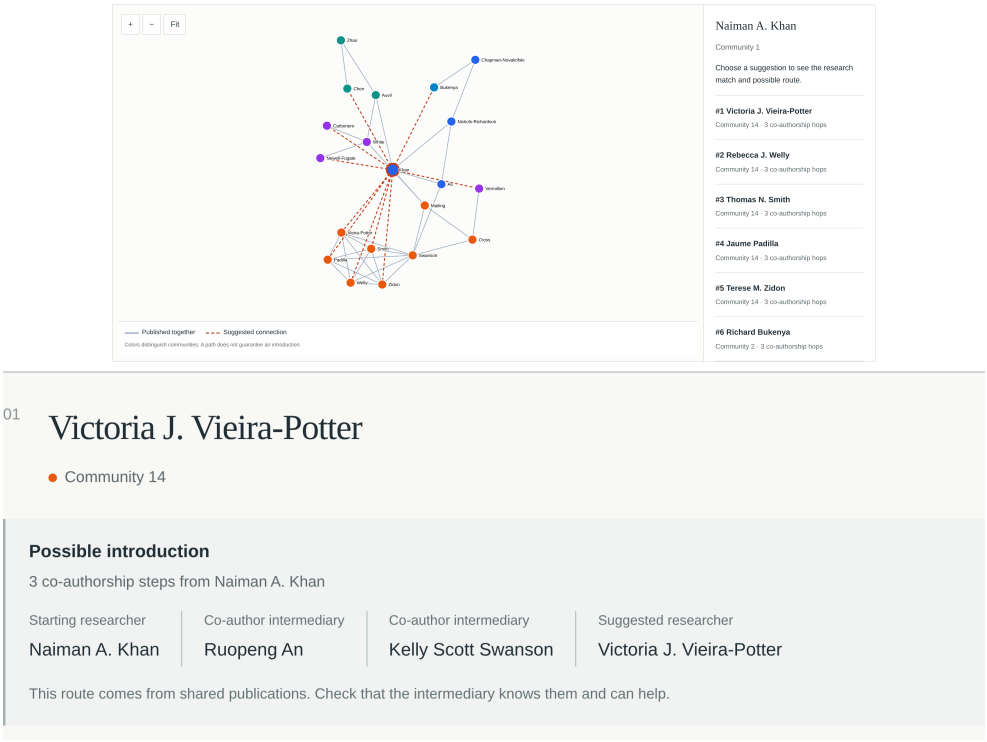



**Figure 1.** The upper image shows Naiman A. Khan's map. Solid lines represent observed co-authorships, and orange dashed lines represent suggestions. The lower image shows a detail of Victoria J. Vieira-Potter's card, with a route through Ruopeng An and Kelly Scott Swanson. Topics and papers appear below this detail. The route is observed in publications, not a confirmed introduction.

**Readable map.** The graph showed too many nodes and was difficult to follow. We narrowed the display to the selected researcher, suggestions, and people along their routes. Community colors help distinguish groups without showing the entire campus graph.

**Correct labels.** Some tags in the original design did not match the researchers shown. We corrected the profile labeling and distinguish inferred research areas from collaboration groups. These labels do not claim verified departmental appointments.

The new entry instructions and card definition have not yet been retested with students. The observations explain why we made the revisions. Further testing is needed to find out whether they help.

# 6 Evaluation

## 6.1 Historical prediction

We used 2015–2019 papers to build the graph and topic profiles, then checked whether suggestions became new cross-community co-authors in 2020–2024. We evaluated 3,492 starting researchers in this dataset. Each had at least one new cross-community tie between training-graph authors. Researchers without such future ties are excluded.

We compare with **Topic-Only** on the same candidate pool and **Triadic closure**, the shared-contact baseline in our proposal. We rank these candidates by their number of shared co-authors, with no additional weighting. These candidates are two hops away. Restricting that baseline to our three-or-more-hop pool would give only zero scores. All methods predict the same future ties, so this compares discovery strategies with different pools.

We measure the fraction of each researcher's top ten suggestions that became collaborators in the later period. This is precision at ten, or **P@10**. It matches the proposal's test of whether the shortlist includes people who later worked together. We average the fraction across starting researchers.

**Table 2.** Prediction of new 2020–2024 cross-community ties. Higher is better.

| Method | P@10 |
|---|---|
| Topic-Only | 0.0079 |
| Triadic closure | 0.0287 |
| BridgeFinder | 0.0131 |

Table 2 shows that BridgeFinder predicts more later ties than Topic-Only (0.0131 vs 0.0079), but fewer than Triadic closure (0.0287). It therefore does not meet the proposal's goal of outperforming both baselines. We have not estimated uncertainty, and these comparisons do not isolate distance from scoring. The student study will test whether the suggestions are useful for discovery despite lower historical prediction than triadic closure.

### 6.2 Planned student study

We will recruit 15 students, aiming for three each from CS, ECE, Bioengineering, Crop Sciences, and Information Sciences. This adds backgrounds beyond those in our first feedback round, but the study remains exploratory.

Each student will choose a starting researcher, open a suggestion and outreach draft, and explain the route and dashed line without coaching. A facilitator will record task completion and whether the student identifies the intermediaries and distinguishes suggestions from existing ties.

For the card comparison, students will choose one of five starting researchers and rate up to 15 shuffled cards, with five from each method above. Method labels are hidden. Shared candidates appear once and count for every method that selected them. We will collect the following measures.

- **Relevant discovery.** Students will rate relevance to the starting researcher's work and novelty to themselves on scales from 1 to 7. Higher ratings mean a better research match or a less familiar suggestion. For BridgeFinder, we target means of at least 5 and a 25-percentage-point gain over triadic closure in cards rated at least 5 on both. Topic-Only is a second comparison.
- **Willingness to contact.** Students will rate how realistic contact feels given the route on a scale from 1 to 7. This checks whether they see a practical next step. For BridgeFinder, we target a mean of at least 5 and improvement over Topic-Only. Because the methods suggest different people, this rating cannot isolate the effect of showing a path.
- **Usability.** At least 12 of 15 students should complete the task and explain the line types correctly without help. The ten-item System Usability Scale will assess ease of using the interactive app, with a target mean of 80/100.

We will compare methods using participant averages, not treat individual cards as independent responses. The rating and analysis tools are implemented. We have not yet run the study.

## 7 Limitations and Next Steps

Co-authorship cannot confirm an introduction or mentoring availability. Our dataset omits external and informal ties, and OpenAlex does not capture every publication. The demo offers 50 starting researchers, and the card study offers five, so some students may not find a suitable starting point. A researcher's topics may also be a poor proxy for a student's interests.

In November, we will run the student study to assess the suggestions and revise the interface. Any ranking changes will use separate validation data.

## Contributions and AI Use

Evan Lin worked on data collection, prediction tests, and the map. Eric Liu worked on scoring, citation analysis, study tools, and formative sessions. Both contributed to the report. Claude assisted with early development and drafting. OpenAI Codex assisted with checks, tests, and revision.

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